

used—this makes for extreme precision. The polarisation of a light beam is twisted when it bounces off a magnetised surface. When this is passed through a polarising filter, the intensity of the beam gives the magnetic alignment of the part of the surface that it touched—this is how the disk is read. For writing, the same laser heats a dot on the disk to a temperature at which the material's magnetic properties can be changed—to reflect either a one or a zero—using a magnetic coil.

Essentially, since a laser is so precise, it can focus on areas much smaller than are used to represent a bit on a traditional hard disk. This makes for higher data densities.

AFC media

In 2001, IBM demonstrated yet another way to get round the SPE: a three-atom-thick layer of ruthenium, sandwiched between two magnetic layers. Known as Anti-Ferromagnetically-Coupled (AFC) media, the multi-layer coating permits disks to store 100 gigabits per square inch, as of now. The ruthenium layer makes the adjacent layers orient themselves magnetically in opposite directions. When bits are written onto layers, they can be placed closer together than if a single layer were used—meaning higher areal density.

Patterned-media recording

One bit on a disk is stored as a group of several small grains. Each of these grains is thermally unstable, and the SPE comes into play. In patterned-media recording, one bit is stored by one grain; a grain can, therefore, afford to be larger. Thus, data integrity can be maintained at very high densities, ranging from 100 gigabits per square inch to tens of terabits per square inch.



Jargon Buster

- **Holographic Storage:** Holograms, unlike, say, regular cameras, record not only the intensity and colour of light, but also its direction. As a result, holographic crystals record data in three dimensions. This means that data is stored over the volume of the crystal, not its surface. Holographic storage has promised storage densities of one terabyte per cubic cm.
- **Areal density:** The areal density refers to how densely bits are packed on the surface of a material or device, for example, a hard disk platter.
- **Cantilever:** A cantilever is a horizontal

fixture, such as a mechanical arm, that projects beyond a vertical support, and is unsupported at one end.

■ **Nanocrystals:** Nanometre-sized semiconductor particles. Due to their size, they don't behave like bulk materials, but like atom-sized quantum entities.

■ **Pipelined operation:** Say an instruction has three parts to it: fetch, read data and execute. A processing unit is doing pipelined operation if, for example, it routinely does the 'read data' part of instruction #2 while doing the 'fetch' part of instruction #3, like in an assembly line.

Patterned media refers to ultra-high density magnetic storage media consisting of arrays of discrete, lithographically patterned magnetic elements, each of which can store one bit of data. Each discrete element is isolated from other elements, but inside each one, polycrystalline grains are strongly coupled, behaving more like a larger, single magnetic grain. Because the SPE limit applies to the entire bit, not to each of the grains, the single bit in patterned media can be much larger than the one in conventional media. The problem with patterned media is that there are no cheap mass-manufacturing procedures.

Probe storage: IBM's Millipede

The tip at the end of a cantilever makes for the ultimate in local confinement of interaction. This makes tip-based storage technologies natural candidates for extending the physical limits being approached by conventional magnetic storage. Probe storage technology makes indentations 10 nm (nano metres) in diameter on a plastic film. Each inden-

tation is a bit, which can be heated back to its original shape. Thousands of tiny probes—tips at the end of a cantilever—move back and forth over the film. Because of the many probes ('feet'), IBM's codename for this technology is 'Millipede'. There are problems inherent in this idea, and IBM's solution was to use micro-electro-mechanical systems (MEMS)-based arrays of cantilevers operating in parallel, with each cantilever performing write, read and erase operations in an individual storage field. Probe storage is expected to use just about as much power as a flash card, but store up to 100 times as many bits as traditional disks.

Flash memory

Companies need to keep shrinking Flash memory as portable devices get smaller—cell phones and digital cameras, for example. Flash sticks transfer data fast and, like hard disks, don't need to be powered on to retain data. These sticks don't have moving parts, so they're much more rugged.

